

Hints & Solutions

#MathBoleTohMathonGo

Q1 (2) - Integration By Substitution

$$I = \int \frac{dx}{e^x + 4e^{-x}} = f(x) + c$$

$$\Rightarrow I = \int \frac{e^x dx}{e^{2x} + 4}$$

$$\text{Let } e^x = t \Rightarrow e^x dx = dt$$

$$\Rightarrow I = \int \frac{dt}{t^2 + 4} = \frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) + c = \frac{1}{2} \tan^{-1}\left(\frac{e^x}{2}\right) + c$$

Hence (2) is the correct answer.

Q2 (4) - Integration By Substitution

$$\text{Let } I = \int \frac{(1+x)}{x(1+xe^x)^2} dx = \int \frac{(1+x)e^x}{(xe^x)(1+xe^x)^2} dx \quad (1 + xe^x = p, e^x(1+x)dx = dp)$$

$$I = \int \frac{dp}{(p-1)p^2}$$

$$\frac{1}{(p-1)p^2} = \frac{A}{(p-1)} + \frac{B}{p} + \frac{C}{p^2}$$

$$1 = Ap^2 + B(p)(p-1) + C(p-1)$$

For $p = 1$, $p = 0$, and $p = -1$, $A = 1$, $C = -1$ and $B = -1$.

$$\Rightarrow 1 = \int \frac{1}{(p-1)} dp - \int \frac{dp}{p} - \int \frac{dp}{p^2} = \ln \frac{(p-1)}{p} + \frac{1}{p} + c = \ln\left(\frac{xe^x}{1+xe^x}\right) + \left(\frac{1}{1+xe^x}\right) + c$$

Hence (4) is the correct answer.

Q3 (3) - Integration By Substitution

$$\begin{aligned} \int \frac{2 \cos \frac{9x}{2} \cos \frac{x}{2} \cos \frac{3x}{2}}{\left[1 - 2\left(2 \cos^2 \frac{3x}{2} - 1\right)\right] \cos \frac{3x}{2}} dx &= \int \frac{2 \cos \frac{9x}{2} \cos \frac{x}{2} \cos \frac{3x}{2}}{3 \cos \frac{3x}{2} - 4 \cos^3 \frac{3x}{2}} dx \\ &= \int \frac{2 \cos \frac{9x}{2} \cos \frac{x}{2} \cos \frac{3x}{2}}{-\cos \frac{9x}{2}} dx = - \int (\cos 2x + \cos x) dx \\ &= -\frac{\sin 2x}{2} - \sin x + c \end{aligned}$$

Hence (3) is the correct answer.

Q4 (2) - Integration By Substitution

$$\text{Put } xe^x = t \therefore I = \int \sec^2 t dt = \tan t + c = \tan(xe^x) + c$$

Hence (2) is the correct answer.

Q5 (2) - Integration By Substitution

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$$I = \int \tan^2 2x \tan 2x \sec 2x dx = \int (\sec^2 2x - 1) \sec 2x \tan 2x dx$$

$$\text{Put } \sec 2x = t \quad \therefore 2 \sec 2x \tan 2x dx = dt$$

$$\therefore I = \frac{1}{2} \int (t^2 - 1) dt = \frac{1}{2} \left(\frac{t^3}{3} - t \right) = \frac{1}{6} (\sec^3 2x - 3 \sec 2x)$$

Hence (2) is the correct answer.

Q6 (3) - Integration By Substitution

$$I = \int \frac{\log\left(\frac{x+1}{x}\right)}{\frac{(x+1)}{x}} \cdot \frac{1}{x^2} dx$$

$$\text{Now put } t = \frac{x+1}{x} = 1 + \frac{1}{x} \quad \therefore dt = -\frac{1}{x^2} dx$$

$$\therefore I = - \int \log t \cdot \frac{1}{t} dt = -\frac{1}{2} (\log t)^2 + c = -\frac{1}{2} \log \left[\left(\frac{x+1}{x} \right)^2 \right] + c$$

Hence (3) is the correct answer.

Q7 (1) - Integration By Substitution

$$I = \int e^{\sin x} \cos x dx = e^{\sin x}$$

$$\text{as } 1 - \sin 2x = (\sin x - \cos x)^2$$

$$\text{Hence } \int e^{\sin x} \cos x dx$$

$$\text{Put } \sin x = t \Rightarrow \cos x dx = dt$$

$$\int e^t dt = e^t + c = e^{\sin x} + c$$

Hence (1) is the correct answer.

Q8 (4) - Integration By Substitution

$$I = \int \frac{2 \cos^2 x - 1}{\cos x} dx$$

$$I = \int (2 \cos x - \sec x) dx = 2 \int \cos x dx - \int \sec x dx$$

$$= 2 \sin x - \log |\sec x + \tan x| + c$$

Hence (4) is the correct answer.

Q9 (2) - Integration By Substitution

$$e^{\log_5 x} = x^{\log_5 e} \text{ (by property of exponential function)}$$

$$\int e^{\log_5 x} dx = \int x^{\log_5 e} dx = \frac{x^{\log_5 5e + 1}}{\log_5 5e + 1} = \frac{x^{\log_5 5e}}{\log_5 5e}$$

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Hence (2) is the correct answer.

Q10 (1) - Integration By Substitution

Here we have $\cos x$ but its d.c.i.e., $-\sin x$ is not present in the numerator and as such we cannot make the substitution of $\cos x = t$. but we simply put $\cos x = t$ to split the integrand into partial fractions.

$$\begin{aligned}\frac{1-\cos x}{\cos x(1+\cos x)} &= \frac{1-t}{t(1+t)} \\ &= \left(\frac{1}{t} - \frac{2}{1+t} \right) = \left(\frac{1}{\cos x} - \frac{2}{1+\cos x} \right) \\ \therefore I &= \int \left(\frac{1}{\cos x} - \frac{2}{1+\cos x} \right) dx = \int \left(\sec x - \sec^2 \frac{x}{2} \right) dx \\ &= \log(\sec x + \tan x) - 2 \tan \left(\frac{x}{2} \right)\end{aligned}$$

Hence (1) is the correct answer.

Q11 (3) - Integration By Substitution

$$\begin{aligned}\text{Put } (1-x) &= t \\ -dx &= dt \\ \int -dt^{23}(1-t) &= \int (t^{24} - t^{23}) dt \\ &= \frac{t^{25}}{25} - \frac{t^{24}}{24} + c = \frac{(1-x)^{25}}{25} - \frac{(1-x)^{24}}{24} + c.\end{aligned}$$

Hence (3) is the correct answer.

Q12 (3) - Integration By Substitution

$$\text{Let } I = \int \frac{dx}{\cos^6 x + \sin^6 x} = \int \frac{\sec^6 x}{1 + \tan^6 x} dx = \int \frac{(1 + \tan^2 x)^2 \sec^2 x dx}{1 + \tan^6 x}$$

If $\tan x = p$, then $\sec^2 x dx = dp$

$$= \int \frac{(1+p^2)^2 dp}{1+p^6} = \int \frac{(1+p^2)}{p^4 - p^2 + 1} dp = \int \frac{p^2 \left(1 + \frac{1}{p^2} \right)}{p^2 \left(p^2 + \frac{1}{p^2} - 1 \right)} dp$$

$$\therefore I = \int \frac{dk}{k^2 + 1} = \tan^{-1}(k) + c$$

If $p - \frac{1}{p} = k$, then $\left(1 + \frac{1}{p^2} \right) dp = dk$

$$= \tan^{-1} \left(p - \frac{1}{p} \right) + c = \tan^{-1}(\tan x - \cot x) + c = \tan^{-1}(-2 \cot 2x) + c$$

Hence (3) is the correct answers.

Q13 (3) - Integration By Substitution

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$$e^{\log x} = x$$

$$1 = \int x \cos x dx + \int \sin x \cos x dx = x \sin x + \cos x - \frac{\cos^2 x}{2} + c$$

Hence (3) is the correct answer.

Q14 (1) - Integration By Substitution

$$I = \int \frac{x + \sin x}{1 + \cos x} dx = \frac{1}{2} \int \sec^2 \frac{x}{2} (x + \sin x) dx$$

$$I = \frac{1}{2} \int x \sec^2 \frac{x}{2} + \frac{1}{2} \int \tan \frac{x}{2} dx$$

$$= \frac{1}{2} x \frac{\tan x/2}{1/2} - \frac{1}{2} \int \frac{\tan x/2}{1/2} + \int \tan x/2 dx = x \tan x/2 + c$$

Hence (1) is the correct answer.

Q15 (2) - Integration By Substitution

$$\text{Let } I = \int x^x (1 + \ln x) dx$$

$$\text{Let } x^x = t \Rightarrow x \ln x = \ln t$$

$$\Rightarrow \left(x \cdot \frac{1}{x} + \ln x \cdot 1 \right) = \frac{1}{t} \frac{dt}{dx}$$

$$\Rightarrow dx(1 + \ln x)x^x = dt$$

$$\therefore I = \int dt \Rightarrow t + c$$

$$I = x^x + c$$

Hence (B) is the correct answer.

Q16 (1) - Integration By Substitution

$$\text{Let } I = \int \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$$

$$\text{Let } t = \sin x - \cos x \Rightarrow t^2 = 1 - \sin 2x$$

$$\therefore \sin 2x = (1 - t^2)$$

$$\therefore I = \int \frac{dt}{9 + 16(1 - t^2)} = \int \frac{dt}{25 - 16t^2}$$

$$= \frac{1}{16} \int \frac{dt}{\left(\frac{5}{4}\right)^2 - t^2} = \frac{1}{16} \times \frac{1}{2\left(\frac{5}{4}\right)} \ln \left| \frac{\frac{5}{4} + t}{\frac{5}{4} - t} \right| + c$$

$$= \frac{1}{40} \ln \left| \frac{5 + 4t}{5 - 4t} \right| + c = \frac{1}{40} \ln \left| \frac{5 + 4(\sin x - \cos x)}{5 - 4(\sin x - \cos x)} \right| + c$$

Hence (1) is the correct answer.

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Q17 (4) - Integration By Substitution

$$\text{Let } I = \int \frac{\left(\sqrt{\frac{x^2+1}{x^2}}\right) \ln\left(\frac{x^2+1}{x^2}\right)}{x^3} dx$$

$$\text{Let } \frac{x^2+1}{x^2} = t$$

$$\therefore -\frac{2}{x^3} dx = dt$$

$$\Rightarrow I = -\frac{1}{2} \int \sqrt{t} \ln t dt$$

$$= -\frac{1}{2} \left[(\ln t) \cdot \frac{2t^{3/2}}{3} - \frac{2}{3} \int \frac{1}{t} \cdot t^{3/2} dt \right]$$

$$= \frac{1}{9} t^{3/2} [2 - 3 \ln t] + c$$

$$= \frac{1}{9} \frac{(x^2+1)\sqrt{x^2+1}}{x^3} \left[2 - 3 \ln\left(\frac{x^2+1}{x^2}\right) \right] + c$$

Hence (4) is the correct answer.

Q18 (3) - Integration By Substitution

$$\int \frac{\cos 2x}{\cos x + \sin x} dx = \int \frac{(\cos x + \sin x)(\cos x - \sin x)}{\cos x + \sin x} dx = \sin x + \cos x + c$$

Hence (3) is the correct answer.

Q19 (1) - Integration By Substitution

$$\text{Let } I = \int \frac{e^{x-1} + x^{e-1}}{e^x + x^e} dx$$

$$\text{Let } e^x + x^e = t \Rightarrow (e^x + e \cdot x^{e-1}) dx = dt$$

$$\Rightarrow e (e^{x-1} + x^{e-1}) dx = dt$$

$$\therefore I = \frac{1}{e} \int \frac{dt}{t} = \frac{1}{e} \ln(e^x + x^e) + c$$

Hence (1) is the correct answer.

Q20 (1) - Integration By Parts

$$I = \text{real part of } \int x e^{(1+i)x} dx$$

$$= \frac{x e^{(1+i)x}}{1+i} - \int \frac{e^{(1+i)x}}{1+i} dx = \frac{x e^{(1+i)x}}{1+i} - \frac{e^{(1+i)x}}{(1+i)^2}$$

$$= e^{(1+i)x} \left[\frac{x(1+i)-1}{(1+i)^2} \right]$$

$$= e^x (\cos x + i \sin x) \left[\frac{(x-1)+ix}{1+i-1} \right]$$

$$= \frac{e^x}{-2} [i \cos x - \sin x] [(x-1) + ix]$$

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$$I = \frac{e^x}{2} [(1-x) \sin x - x \cos x] + c$$

Hence (1) is the correct answer.

Q21 (3) - Integration By Parts

By parts

$$I = -x^2 \cos x + 2x \sin x + 2 \cos x + c \text{ with alternate } +, - \text{ sign.}$$

Differentiation	Integration
x^2	$\sin x$
$2x$	$-\cos x$
2	$-\sin x$
0	$\cos x$

Hence (3) is the correct answer.

Q22 (2) - Integration By Parts

$$\text{Let } I = \int \frac{x \cos x}{(x \sin x + \cos x)^2} dx$$

$$\text{Let } \frac{1}{x \sin x + \cos x} = t \Rightarrow \frac{(x \sin x + \cos x) \cdot 0 - 1(x \cos x + \sin x - \sin x)}{(x \sin x + \cos x)^2} = \frac{dt}{dx}$$

$$\Rightarrow \frac{-x \cos x}{(x \sin x + \cos x)^2} = \frac{dt}{dx}$$

$$\therefore I = - \int dt = -\frac{1}{x \sin x + \cos x} + c.$$

Hence (2) is the correct answer.

Q23 (1) - Integration By Parts

$$I = \int e^p (\sec^2 p + \tan p) dp$$

$$(\text{If } p = \tan^{-1} x \Rightarrow x = \tan p \Rightarrow dx = \sec^2 p dp)$$

$$= e^p \tan p = x e^{\tan^{-1} x} + c$$

Hence (1) is the correct answer.

Q24 (4) - Integration By Parts

$$\int \frac{\sqrt{\cot x}}{\sin x \cos x} dx = \int \frac{\sqrt{\cot x}}{\cot x} \cdot \operatorname{cosec}^2 x dx = \int \frac{\operatorname{cosec}^2 x}{\sqrt{\cot x}} dx = -2\sqrt{\cot x} + B$$

$$\text{Hence } A = -2$$

Hence (4) is the correct answer.

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Q25 (3) - Integration By Parts

$$\text{Let } I = \int \frac{\ln(x/e)}{(\ln x)^2} dx = \int \frac{\ln(x)-1}{(\ln x)^2} dx$$

$$\text{Put } \ln x = t \Rightarrow x = e^t \Rightarrow dx = e^t dt$$

$$I = \int e^t \left(\frac{t-1}{t^2} \right) dt = \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt = \frac{e^t}{t} + c = \frac{x}{\ln x} + c$$

Hence (3) is the correct answer.

Q26 (1, 2) - Integration By Parts

$$\text{Put } t = \sin^2 x$$

The integral reduces to

$$I = \frac{1}{2} \int e^t (2-t) dt = \frac{3}{2} e^t - \frac{te^t}{2} + c$$

$$= \frac{1}{2} e^{\sin^2 x} (3 - \sin^2 x) + c \quad (\text{option 1})$$

$$= e^{\sin^2 x} \left(1 + \frac{1}{2} \cos^2 x \right) + c \quad (\text{option 2})$$

Hence (1) & (2) are the correct answers.

Q27 (1) - Special Substitutions and Integrals

Put $x = a \cos^2 \theta + b \sin^2 \theta$ the given integral becomes

$$I = \int \frac{2(b-a) \sin \theta \cos \theta d\theta}{\{(a \cos^2 \theta + b \sin^2 \theta - a)(a \cos^2 \theta + b \sin^2 \theta - b)\}^{\frac{1}{2}}}$$

$$= \frac{2(b-a) \sin \theta \cos \theta d\theta}{(b-a) \sin \theta \cos \theta} = \left(\frac{b-a}{b-a} \right) \int 2d\theta = 2\theta + c = 2 \sin^{-1} \sqrt{\left(\frac{x-a}{b-a} \right)} + c$$

Hence (1) is the correct answer.

Q28 (2) - Special Substitutions and Integrals

Let $\sin x = A(\sin x - \cos x) + B \cdot d \cdot c \text{ of } (\sin x - \cos x)$

or $\sin x = A(\sin x - \cos x) + B(\cos x + \sin x)$

or $\sin x = (A+B) \sin x + (B-A) \cos x$

equating the coefficient of $\sin x$ and $\cos x$, we get

$$A+B=1 \text{ and } B-A=0$$

$$A=1/2, B=1/2$$

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$$I = \int \frac{\frac{1}{2}(\sin x - \cos x) + \frac{1}{2}(\cos x + \sin x)}{\sin x - \cos x} dx$$

$$= \frac{1}{2} \int dx + \frac{1}{2} \int \frac{\cos x + \sin x}{\sin x - \cos x} dx = \frac{x}{2} + \frac{1}{2} \log \sin x - \cos x + c$$

Hence (2) is the correct answer.

Q29 (1) - Special Substitutions and Integrals

$$I = \int \frac{(x^4 - x)^{\frac{1}{4}}}{x^5} dx. \text{ Put } 1 - \frac{1}{x^3} = t$$

$$\therefore \frac{3}{x^4} dx = dt, \therefore I = \frac{1}{3} \int t^{\frac{1}{4}} dt$$

$$= \frac{1}{3} \cdot \frac{t^{5/4}}{5/4} + c = \frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{5/4} + c$$

Hence (1) is the correct answer.

Q30 (1) - Special Substitutions and Integrals

$$\text{Let } I = \int \frac{3+2\cos x}{(2+3\cos x)^2} dx$$

Multiplying Nr. & Dr. by $\operatorname{cosec}^2 x$

$$\Rightarrow I = \int \frac{(3 \operatorname{cosec}^2 x + 2 \cot x \operatorname{cosec} x)}{(2 \operatorname{cosec} x + 3 \cot x)^2} dx$$

$$= - \int \frac{-3 \operatorname{cosec}^2 x - 2 \cot x \operatorname{cosec} x}{(2 \operatorname{cosec} x + 3 \cot x)^2} dx$$

$$= \frac{1}{2 \operatorname{cosec} x + 3 \cot x} = \left(\frac{\sin x}{2+3\cos x} \right) + c$$

Hence (1) is the correct answer.

Q31 (1) - Special Substitutions and Integrals

Substitute $\sqrt{2x+1} = p$ and solve by partial fractions.

Hence (1) is the correct answer.

Q32 (1) - Special Substitutions and Integrals

$$\text{Substituting } x = p^6, dx = 6p^5 dp$$

$$I = \int \frac{6p^5(p^6 + p^4 + p)}{p^6(1+p^2)} dp = \int \frac{6(p^5 + p^3 + 1)}{(p^2 + 1)} dp = \int 6p^3 dp + \int \left(\frac{6}{p^2 + 1} \right) dp$$

$$= \frac{6p^4}{4} + 6 \tan^{-1} p = \frac{3}{2} x^{\frac{2}{3}} + 6 \tan^{-1} \left(x^{\frac{1}{6}} \right) + c$$

Hence (1) is the correct answer.

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Q33 (3) - Special Substitutions and Integrals

$$\text{Let } I = \int \sqrt{\tan x} + \sqrt{\cot x} dx$$

$$= \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{2 \sin x \cos x}} dx$$

$$= \sqrt{2} \int \frac{(\sin x + \cos x)}{\sqrt{\sin 2x}} dx$$

$$\text{Let } t = \sin x - \cos x$$

$$\Rightarrow t^2 = 1 - \sin 2x$$

$$\Rightarrow dt = (\cos x + \sin x) dx$$

$$\therefore I = \sqrt{2} \int \frac{dt}{\sqrt{1-t^2}} = \sqrt{2} \sin^{-1}(t) + c$$

$$I = \sqrt{2} \sin^{-1}(\sin x - \cos x) + c$$

Hence (3) is the correct answer.

Q34 (2) - Special Substitutions and Integrals

$$\text{Let } I = \int \frac{1}{x^2(x^4+1)^{3/4}} dx$$

$$= \int \frac{dx}{x^5 \left(1 + \frac{1}{x^4}\right)^{3/4}} dx$$

$$\text{Let } 1 + x^{-4} = t \Rightarrow \frac{-4}{x^5} dx = dt \Rightarrow \frac{1}{x^5} dx = -\frac{1}{4} dt$$

$$\Rightarrow I = -\frac{1}{4} \int \frac{dt}{t^{3/4}} = -\frac{1}{4} \int t^{-3/4} dt = -\frac{1}{4} \times 4t^{1/4} + c$$

$$\Rightarrow I = -\left(1 + \frac{1}{x^4}\right)^{1/4} + c$$

Hence (2) is the correct answer.

Q35 (1) - Special Substitutions and Integrals

$$\text{Let } I = \int \frac{\sqrt{5+x^{10}}}{x^{16}} dx$$

$$\text{Let } 1 + \frac{5}{x^{10}} = t \Rightarrow 5 \left(\frac{-10}{x^{11}} \right) dx = dt$$

$$\therefore \frac{1}{x^{11}} dx = -\frac{1}{50} dt$$

$$\Rightarrow I = -\frac{1}{50} \int \sqrt{t} dt$$

$$= -\frac{1}{50} \times \frac{2}{3} t^{3/2} = -\frac{1}{75} \left(1 + \frac{5}{x^{10}}\right)^{3/2} + c.$$

Hence (1) is the correct answer.

Q36 (1) - Special Substitutions and Integrals

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$$\text{Let } I = \int \frac{dx}{5+4\cos x}$$

$$\text{Let } t = \tan \frac{x}{2} \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\Rightarrow I = \int \frac{\frac{2dt}{1+t^2}}{5+4\left(\frac{1-t^2}{1+t^2}\right)} = 2 \int \frac{dt}{t^2+9}$$

$$= 2 \cdot \frac{1}{3} \cdot \tan^{-1}\left(\frac{t}{3}\right) + c$$

$$\Rightarrow I = \frac{2}{3} \tan^{-1}\left(\frac{1}{3} \tan \frac{x}{2}\right) + c$$

Hence (1) is the correct answer.

Q37 (2) - Special Substitutions and Integrals

$$\text{Let } I = \int \frac{dx}{(2x+3)\sqrt{4x+5}}$$

$$\text{Put } 4x+5 = t \Rightarrow x = \frac{t-5}{4}$$

$$dx = \frac{dt}{4}$$

$$\Rightarrow I = \frac{1}{4} \int \frac{dt}{\left(\frac{2t-10}{4}+3\right)\sqrt{t}} = \frac{1}{2} \int \frac{dt}{(t+1)\sqrt{t}}$$

$$\text{Let } \sqrt{t} = u$$

$$\therefore I = \int \frac{du}{u^2+1} = \tan^{-1} \sqrt{t} + c$$

$$\Rightarrow I = \tan^{-1} \sqrt{4x+5} + c$$

Hence (2) is the correct answer.

Q38 (1) - Special Substitutions and Integrals

$$\text{Let } I = \int e^{ax} \cos bx dx$$

$$= \text{real part of } \int e^{ax} e^{ibx} dx$$

$$= \text{real part of } \int e^{(a+ib)x} dx$$

Hence (1) is the correct answer.

Q39 (2) - Special Substitutions and Integrals

$$I_n = \int (\ln x)^n dx$$

$$I_n = x(\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$I_n = x(\ln x)^n - n I_{n-1}$$

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$$I_n + nI_{n-1} = x(\ln x)^n$$

Hence (2) is the correct answer.

Q40 (2) - Special Substitutions and Integrals

$$I = \int \frac{dx}{x-x^3} = \int \frac{dx}{x^3\left(\frac{1}{x^2}-1\right)}$$

$$\text{Let } \frac{1}{x^2} = t \Rightarrow -\frac{2}{x^3} dx = dt$$

$$\therefore I = -\frac{1}{2} \ln |t-1| + c = -\frac{1}{2} \ln \left| \frac{1-x^2}{x^2} \right| + c$$

$$= \frac{1}{2} \ln \left| \frac{x^2}{1-x^2} \right| + c \Rightarrow A = \frac{1}{2}$$

Hence (2) is the correct answer.